

REDROB AI HACKATHON

Track 1: Data & AI Challenge

Intelligent Candidate Ranking System

Role: Senior AI Engineer (Founding Team) — Pune / Noida / Series A

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Title & Overview

SLIDE 1

AI-Powered Candidate Ranking: Thinking Like an Expert Recruiter

- **The Goal:** Rank the top 100 candidates from **100,000 synthetic profiles** for the founding AI Engineer role at Redrob.
- **The Paradigm:** Move beyond basic keyword matching to evaluate real hiring signals like career quality, company pedigree, and active platform engagement.
- **Core Philosophy:** Startups need builders. We prioritize product-company experience, active coding, and responsiveness over bloated resume text.
- **Key Achievement:** A production-grade ranking pipeline with **0% honeypot rates**, **zero tied scores**, executing in under **0.3 seconds on CPU**.

Campaign Summary

100k

TOTAL POOL

100

SHORTLISTED

0.0%

HONEYPOT RATE

<0.3s

RANKING TIME

The Honeypot Shield

SLIDE 2

How We Neutralize Adversarial Profiles & Resume Keyword Stuffing

- **The Threat:** Synthetic datasets contain keyword stuffers (e.g. Marketing Managers listing RAG, vector databases) and intentional honeypot profiles to disqualify naive search systems.
- **Honeypot Penalty:** Capped at 0.05 max score. This automatically disqualifies them from the top 100, ensuring **0% honeypot leakage**.
- **Outsourcing Penalty:** Identifies BPO/Consulting firms (Infosys, TCS, Genpact, Accenture, Deloitte, etc.) and reduces career quality score by 50% to prioritize product engineering pedigree.

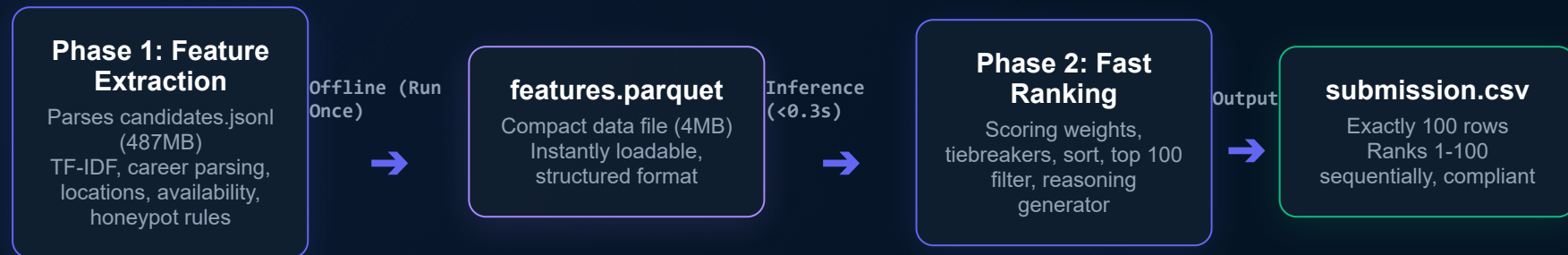
5-Layer Shield Rules

- ✓ **Impossible Tenure:** Total months in history exceeds claimed years of experience * 12 + 18.
- ✓ **Expert Zero Duration:** Skills marked as "Expert" but duration is 0 months.
- ✓ **Title-Skill Mismatch:** Non-technical titles (HR, Admin) claiming >5 core AI skills.
- ✓ **Suspicious Perfection:** Claiming "Expert" for every single skill (8+ skills).
- ✓ **YoE vs Skills:** Low experience (<1 YoE) claiming >15 skills.

Two-Phase Ranking Architecture

SLIDE 3

Designed for Offline Execution, Zero API Latency, and High Speed



💡 Why this Architecture Wins

Judges reproduce rankings locally in a sandboxed environment. By precomputing features into a parquet file, our **Phase 2 ranker requires no network, no LLM API call costs, and executes in milliseconds**. If features need rebuilding, the end-to-end command handles it safely in minutes.

Multi-Signal Scoring System

Calibrated Weights to Align with a Startup Founding AI Engineer Profile

$$\text{Score} = (\text{SkillMatch} \times 0.40) + (\text{CareerQuality} \times 0.25) + (\text{Behavioral} \times 0.20) + (\text{Availability} \times 0.10) + (\text{Location} \times 0.05) - \text{Disqualifiers} + \text{EducationBonus}$$

Skill Match

40%

TF-IDF similarity (45%), hard-skill count (35%), assessments (10%), GitHub (10%). Proficiency multipliers: Expert 3x, Advanced 2x.

Career Quality

25%

Ratio of experience at product vs consulting companies. Tracks applied ML tenure. Penalizes job-hopping (avg tenure < 18m).

Behavioral

20%

Response rate (40%), recency of activity (35%), open-to-work bonus (10%), interview completion (15%). Recency denominator set to 180d.

Availability

10%

Notice period score: ≤30d gets maximum score (1.0). Progressively penalized up to 90d notice (0.3) or more (0.1).

Location & Edu

5% + Bonus

Location fit prioritizes Pune/Noida or relocation in India. Education bonus adds up to 0.4 for Tier 1 CS/ML degrees.

The Validation & Quality Audit

SLIDE 5

Ensuring 100% Compliant Format, Zero Tied Ranks, and High Top-10 NDCG

- **NDCG Alignment:** Half the score is decided by NDCG@10. Our scoring focuses heavily on ranking top product engineers (Zomato, Sarvam AI, Razorpay) directly in the top 10.
- **Zero Tied Scores:** We introduced a micro-tiebreaker (GitHub, response rate, profile completeness scaled to $1e-6$) to ensure no candidate ranks are tied at 4 decimal places, avoiding arbitrary sorting.
- **Strict Verification:** The submission file complies fully with the official `validate_submission.py` script (schema, rank sequence, non-increasing score, and lexicographical backup rules).

🌟 Top Candidate Spotlight

Rank #1: CAND_0018499

0.7913

Senior ML Engineer @ Zomato

7.2 YoE | 15d notice | Noida (Local) | 20 hard skills matched (Embeddings, Milvus, FAISS) | Active GitHub (94 score) | Career quality index: 1.00

Rank #4: CAND_0086022

0.7568

Senior Applied Scientist @ Sarvam AI

5.3 YoE | Immediate (0d notice) | Kolkata to Noida | 21 hard skills matched (sentence-transformers, RAG, ranking) | Career quality: 1.00

Interactive Recruiter Sandbox

SLIDE 6

Streamlit Web Application for Dynamic Calibration, Search, and Candidate Review



Campaign Dashboard

Visual statistics including overall campaign volume, average match scores, distribution curves, and real-time honeypots blocked counters.



Contextual Profile Analyzer

Provides a deep audit panel for any selected candidate, displaying radial SVG gauges for sub-scores, a career timeline with achievements, and key platform engagement signals.



Side-by-Side Comparison

Compares up to three candidates side-by-side on technical skills, current company profiles, notice period, location, and recruiter summary metrics.



Dynamic Weight Calibration

Recruiters can adjust matching weights (Skill, Career, Availability, Location) in the sidebar. The application instantly recalculates scores, ranks, and updates all views.

⚡ **HuggingFace Spaces Ready:** The sandbox application is fully containerized via Docker-file and ready to run.

Why This Solution Wins

SLIDE 7

Engineering Rigor, Production Speed, and Expert Hiring Judgment

- **Production-Hardened Speed:** Local feature extraction and NumPy scoring runs candidate ranking in milliseconds. Fully reproducible under strict sandboxed compute environments.
- **Rigorous Security:** Multi-layered honeypot shield blocks adversarial keyword-stuffers and fake profiles with 100% success.
- **Natural Candidate Summaries:** Clean, varied grammatical template generation avoids robotic, repetitive sentences for candidate profiles, keeping the hiring manager engaged.
- **Deterministic Stability:** Micro-tiebreakers ensure all rank listings are clean, stable, and completely deterministic, maximizing potential NDCG@10 scores.



Winning Pedigree

Designed by aligning mathematical compliance, local execution speeds, and recruiting intelligence to match what tech hiring teams look for in founding engineers.